# LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

## Practice Pack: Structure of the Atom

*Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow*

Name: _____ Roll No.: _____ Date: _____

**Marks obtained:** _____

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**General instructions:**

1. Level 1 checks the ideas, Level 2 is board-style questions, and Level 3 is Foundation-level challenge.
2. Shell capacities: K = 2, L = 8, M = 8 (for the first 18 elements). Mass number = protons + neutrons.

### LEVEL 1: CONCEPTS

**Q1.** Name the three sub-atomic particles and give the charge on each.

**Q2.** What did Rutherford conclude from the observation that most alpha particles passed straight through the gold foil?

**Q3.** Define atomic number and mass number.

**Q4.** What are isotopes? Give one example.

**Q5.** What are isobars? Give one example.

### LEVEL 2: BOARD-STYLE QUESTIONS

**Q6.** Write the electronic configuration and valency of: (a) sodium (Z = 11), (b) oxygen (Z = 8), (c) chlorine (Z = 17), (d) sulphur (Z = 16).

**Q7.** Give three observations of Rutherford's alpha-particle scattering experiment and the conclusion from each.

**Q8.** Chlorine occurs as two isotopes: Cl-35 (75%) and Cl-37 (25%). Find the average atomic mass of chlorine.

**Q9.** Give two uses of isotopes.

**Q10.** Compare the Bohr model with the Rutherford model. What problem in Rutherford's model did Bohr solve?

## LEVEL 3: CHALLENGE (FOUNDATION)

**Q11.** Bromine has two isotopes, Br-79 (49.7%) and Br-81 (50.3%). Calculate its average atomic mass.

**Q12.** The average atomic mass of an element X is 16.2 u. It has two isotopes, X-16 and X-18. Find the percentage of each.

**Q13.** An atom has 2 electrons in its K shell and 6 in its L shell. Identify the element, its valency, and whether it is likely to gain or lose electrons. Explain.

*~ Draw the shells for each element you write about. It makes valency obvious! ~*